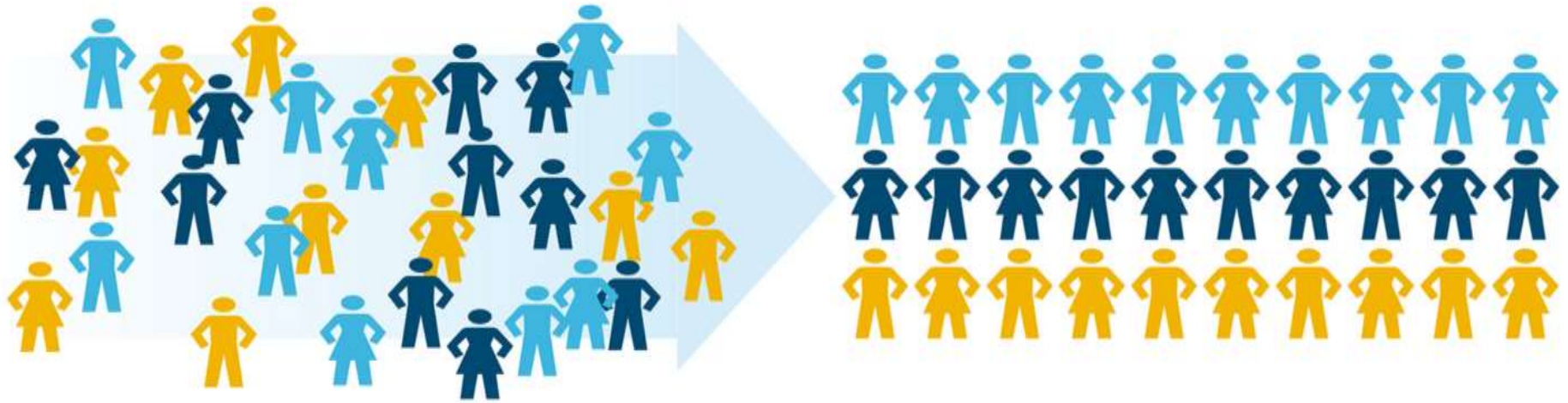


Customer Segmentation ~ RFM Analysis



Compiled By : Mainak Ghosh

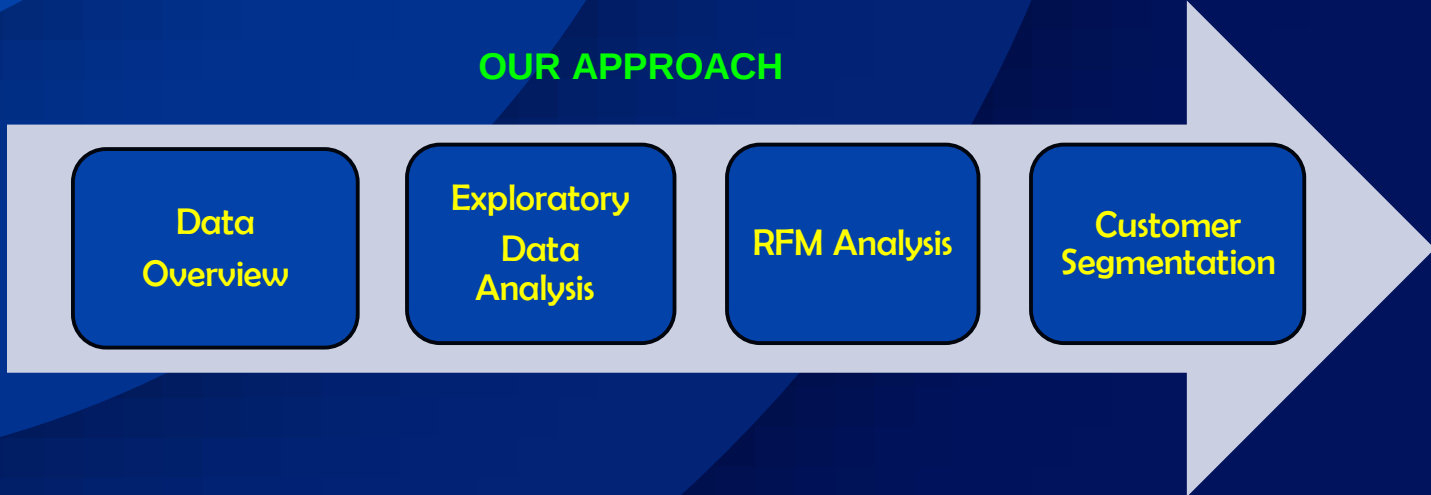
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Problem Statement :

An automobile parts manufacturing company has collected data of transactions for 3 years. They do not have any in-house data science team, thus they have hired you as their consultant. Your job is to use your magical data science skills to provide them with suitable insights about their data and their customers.

OUR APPROACH



About Dataset

The given dataset consists information regarding orders received by the company from different customers for different products and their details. The information is received in form of a table with 2747 rows and 20 columns. But the data received is not as per our requirement. Hence we have tailored the data as per the business prospective. We have deleted some unwanted columns, add new column 'REGENCY' derived by the difference in days from the corresponding order date to the latest order date (31-05-2020) which is considered as our reference date. In the final dataset, there are 2747 rows and 12 columns. **There is no duplicate row present in the dataset. There is no null value too. We have also checked that there is no other type of anomaly (special characters, symbols etc.) in the dataset.**

The sample of final dataset on which we have worked is furnished below:

	QUANTITY ORDERED	PRICE EACH	SALES	ORDER DATE	STATUS	PRODUCT LINE	MSRP	PRODUCT CODE	CUSTOMER NAME	COUNT RY	DEAL SIZE	REGENCY
0	30	95.70	2871.00	2018-02-24	Shipped	Motorcycles	95	S10_1678	Land of Toys Inc.	USA	Small	827 days
1	34	81.35	2765.90	2018-05-07	Shipped	Motorcycles	95	S10_1678	Reims Collectables	France	Small	755 days
2	41	94.74	3884.34	2018-07-01	Shipped	Motorcycles	95	S10_1678	Lyon Souvenirs	France	Medium	700 days
3	45	83.26	3746.70	2018-08-25	Shipped	Motorcycles	95	S10_1678	Toys4Grown Ups.com	USA	Medium	645 days
4	36	96.66	3479.76	2018-10-28	Shipped	Motorcycles	95	S10_1678	Technics Stores Inc.	USA	Medium	581 days

Descriptive Statistics and other related Information

	count	mean	std	min	25%	50%	75%	max	Ske w	Kurt
QUANTITY ORDERED	2747	35.1030	9.7621	6.0	27.0	35.0	43.0	97.0	0.3693	0.4429
PRICEEACH	2747	101.0989	42.0425	26.88	68.745	95.55	127.1	252.87	0.6972	0.2285
SALES	2747	3553.0475	1838.9539	482.13	2204.35	3184.8	4503.095	14082.8	1.1559	1.7731
MSRP	2747	100.6916	40.1148	33.0	68.0	99.0	124.0	214.0	0.5756	- 0.1395
RECECY	2747	383 days	230 days	0 days	196 days	342 days	570 days	876 days	----	----

```
<<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2747 entries, 0 to 2746
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   QUANTITYORDERED  2747 non-null   int64
1   PRICEEACH        2747 non-null   float64
2   SALES            2747 non-null   float64
3   ORDERDATE        2747 non-null   datetime64[ns]
4   STATUS           2747 non-null   object
5   PRODUCTLINE      2747 non-null   object
6   MSRP             2747 non-null   int64
7   PRODUCTCODE      2747 non-null   object
8   CUSTOMERNAME     2747 non-null   object
9   COUNTRY          2747 non-null   object
10  DEALSIZE         2747 non-null   object
11  RECECY           2747 non-null   timedelta64[ns]
dtypes: datetime64[ns](1), float64(2), int64(2), object(6), timedelta64[ns](1)
memory usage: 257.7+ KB
```

Number of duplicate rows = 0

	count	unique	top	freq
STATUS	2747	6	Shipped	2541
PRODUCTLINE	2747	7	Classic Cars	949
PRODUCTCODE	2747	109	S18_3232	51
CUSTOMERNAME	2747	89	Euro Shopping Channel	259
COUNTRY	2747	19	USA	928
DEALSIZE	2747	3	Medium	1349

Exploratory Data Analysis

DATA



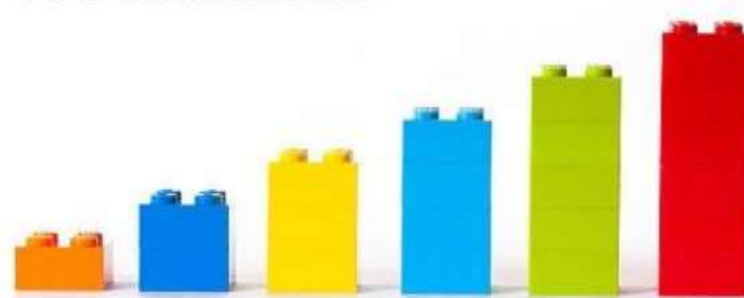
SORTED



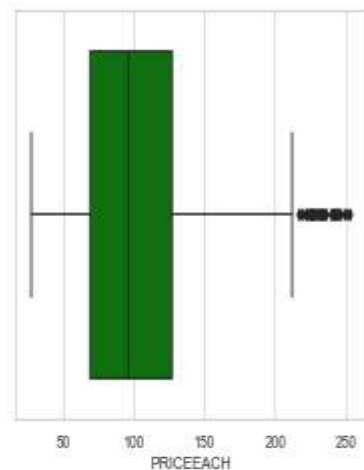
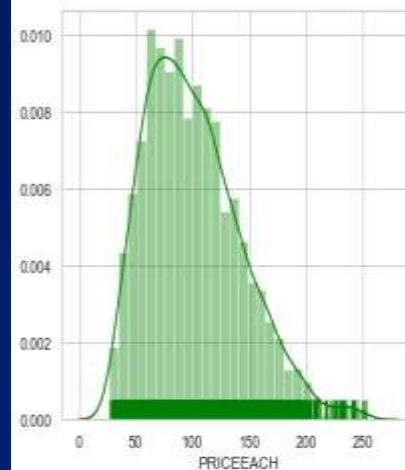
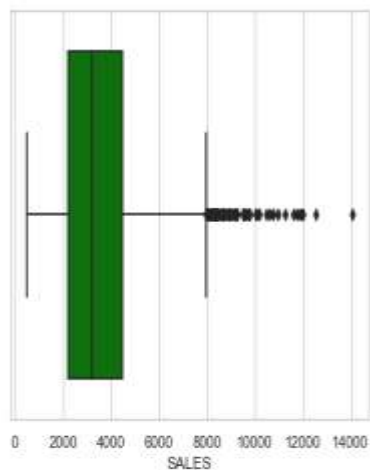
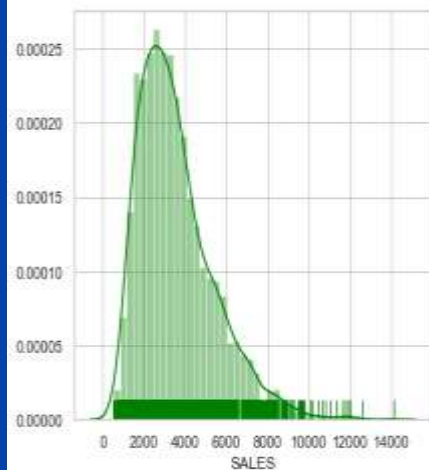
ARRANGED



VISUALIZED

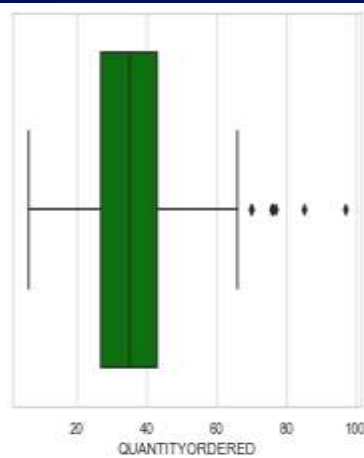
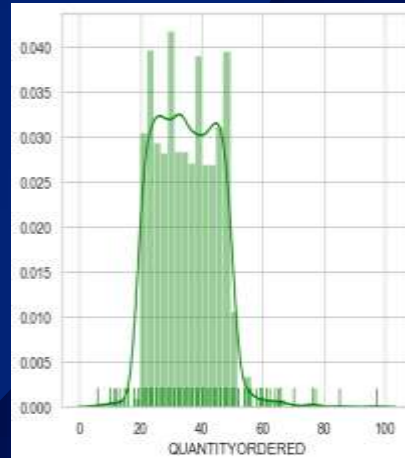
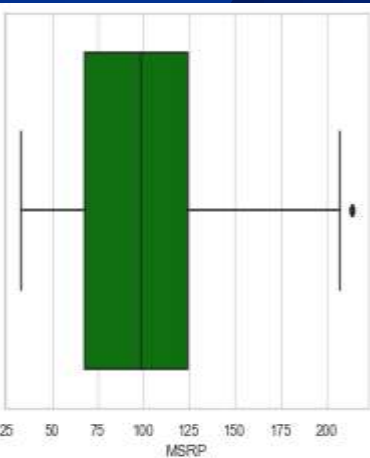
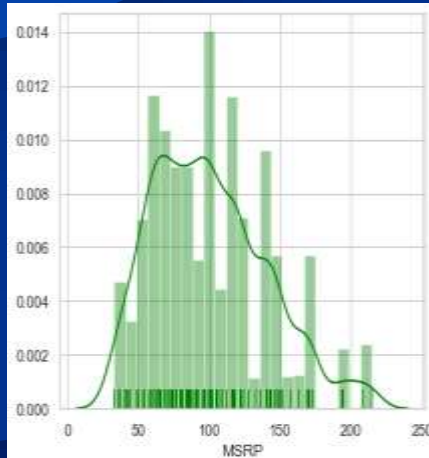


SALES



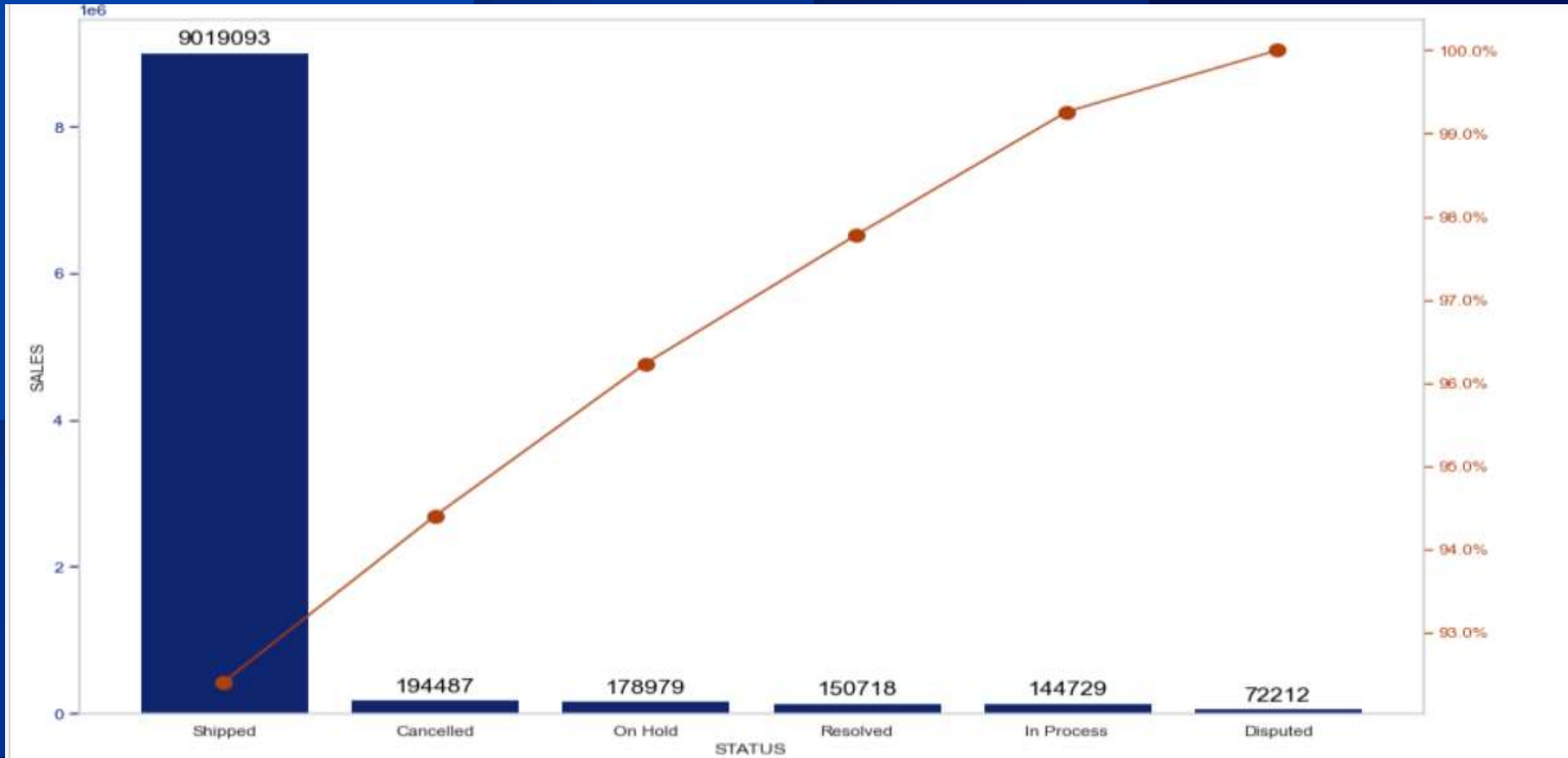
PRICE

MSRP

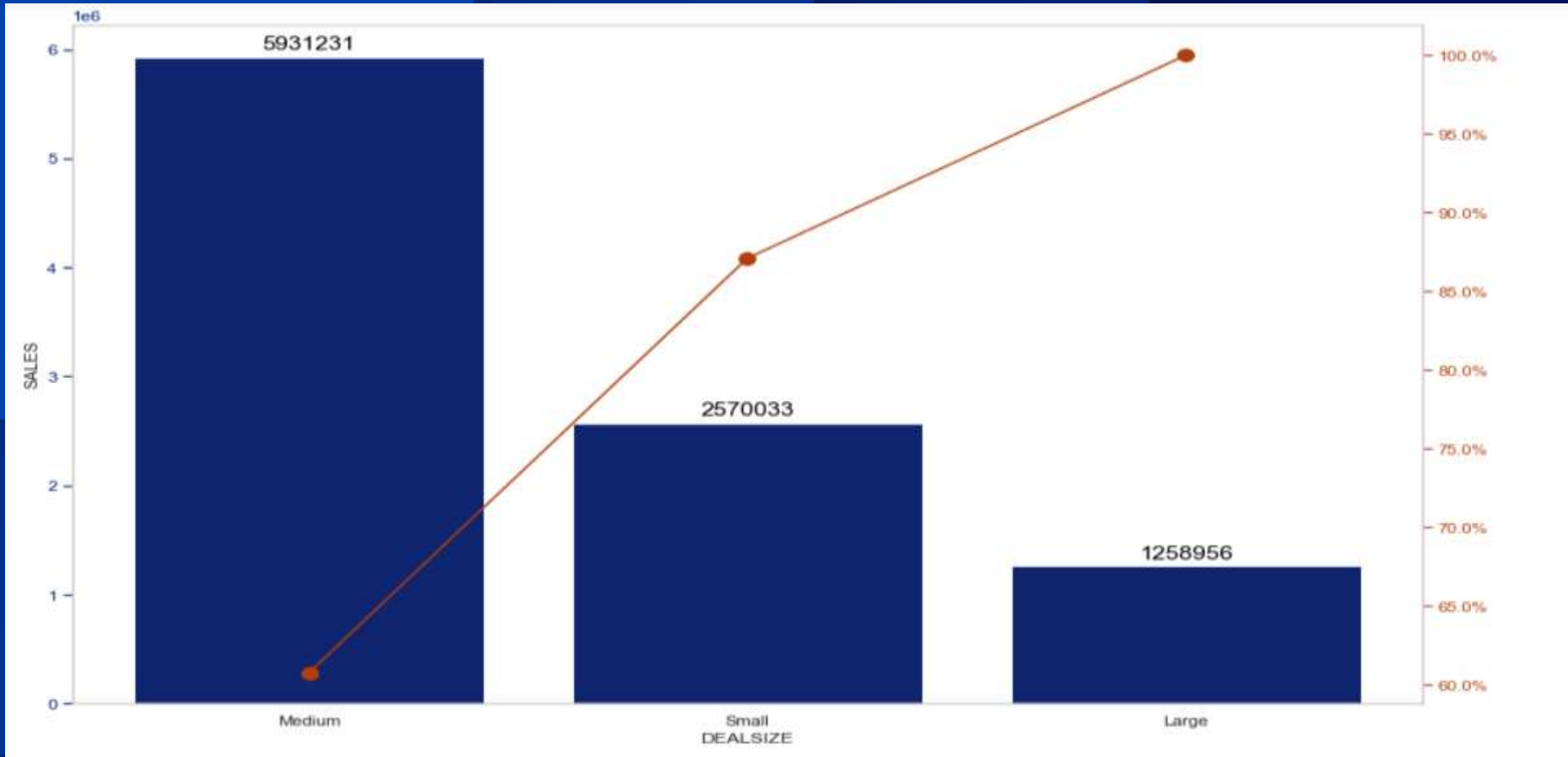


ORDERED

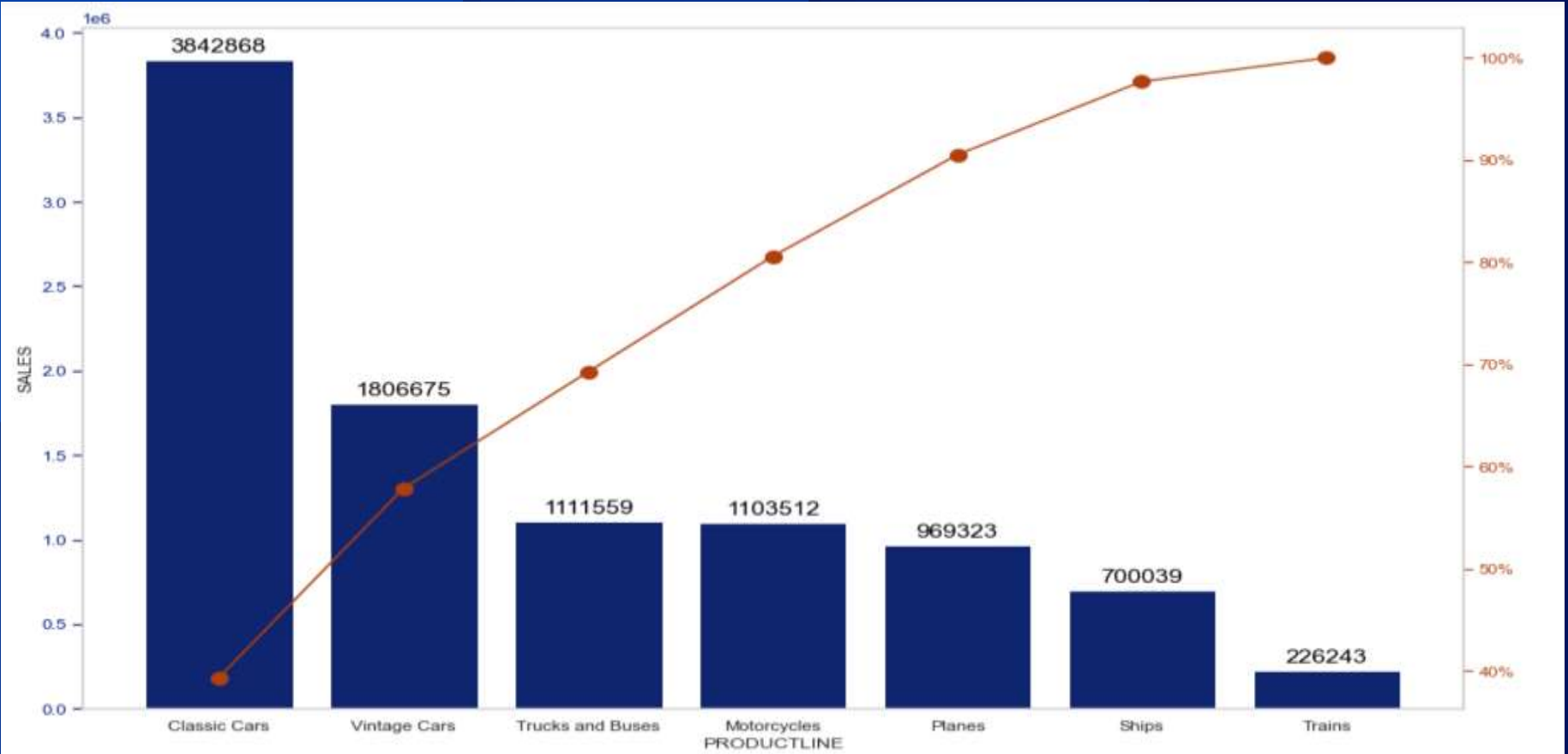
Status wise Sales Distribution



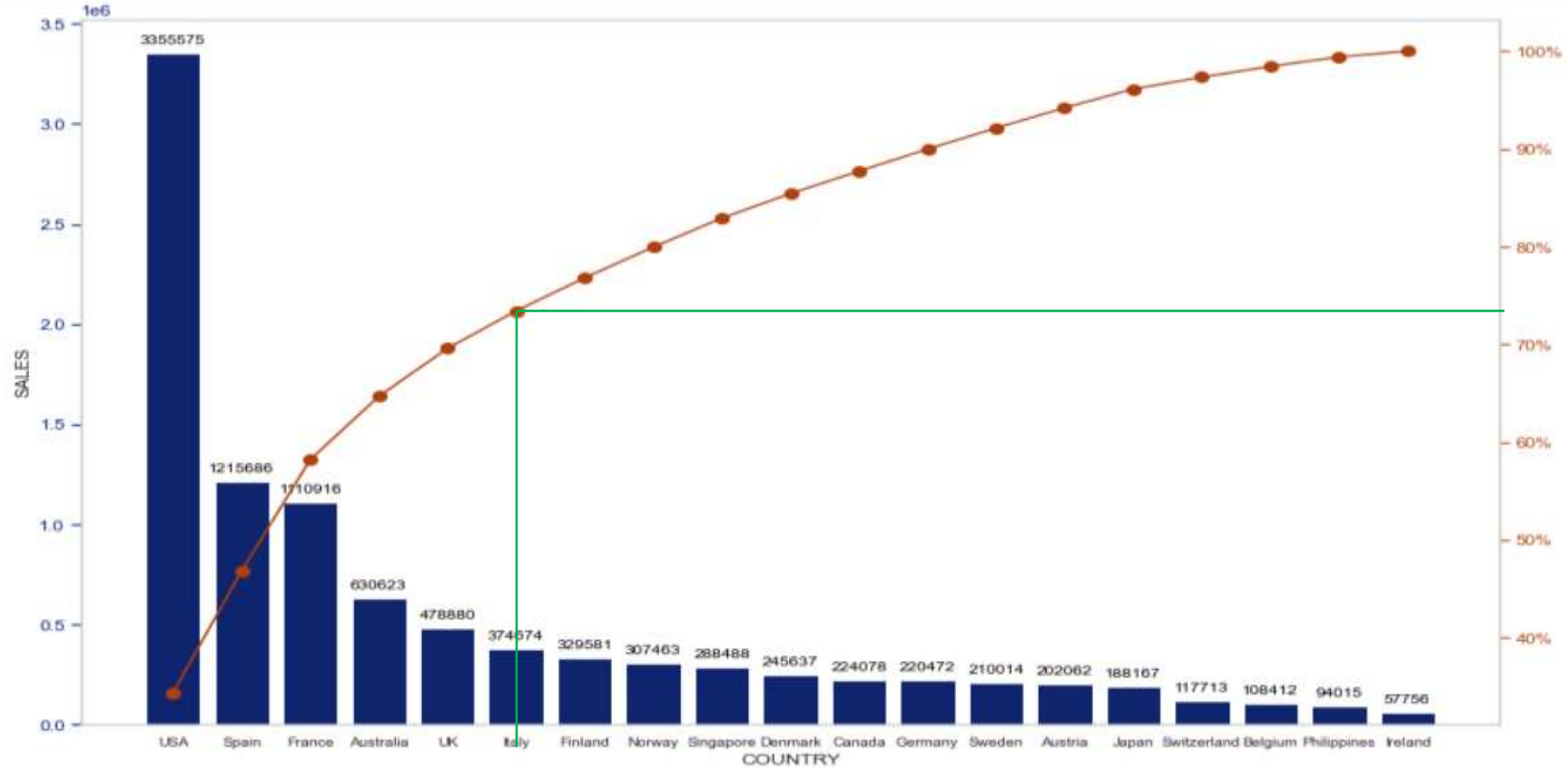
Dealsize wise Sales Distribution



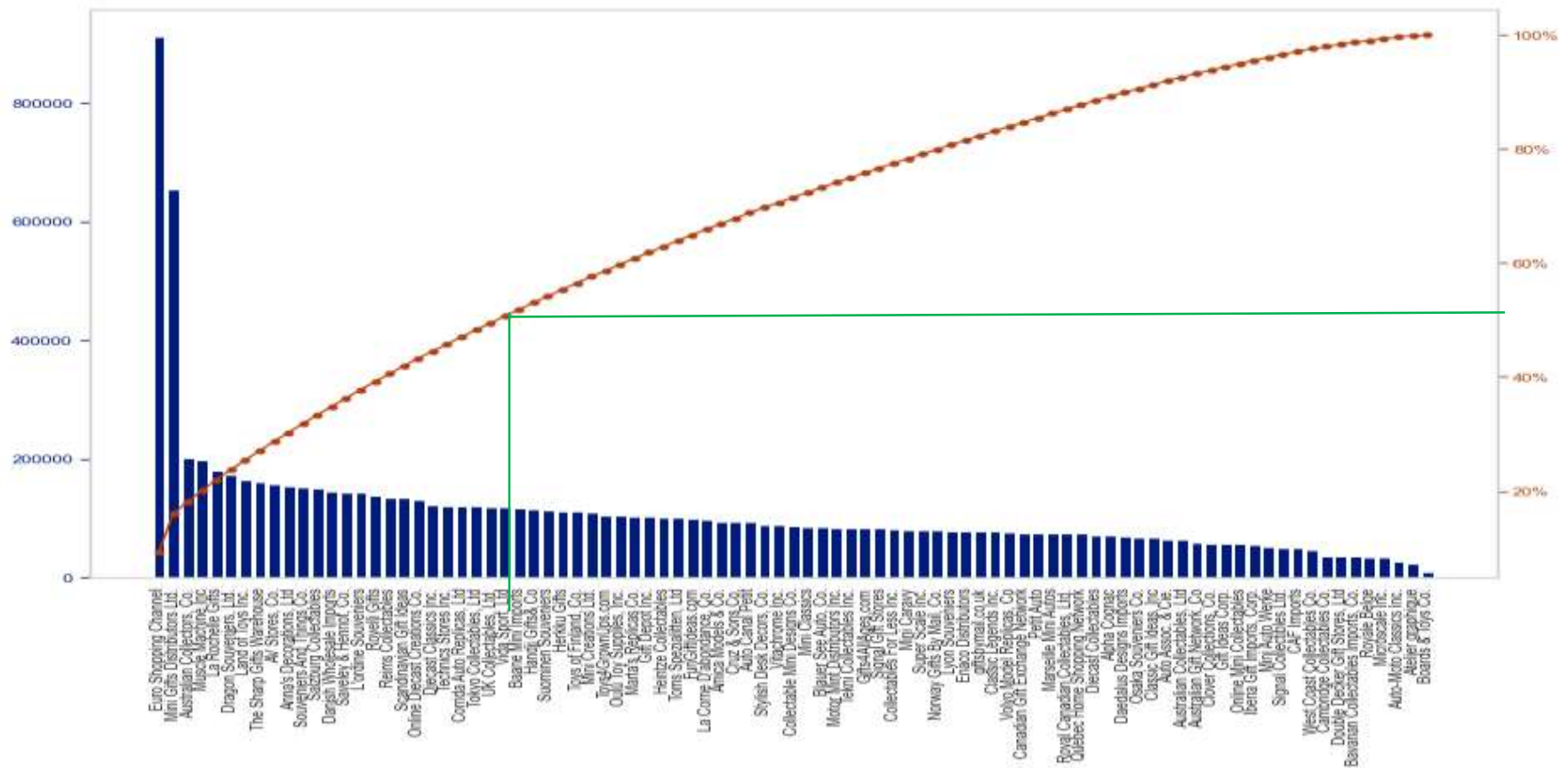
Product wise Sales Distribution

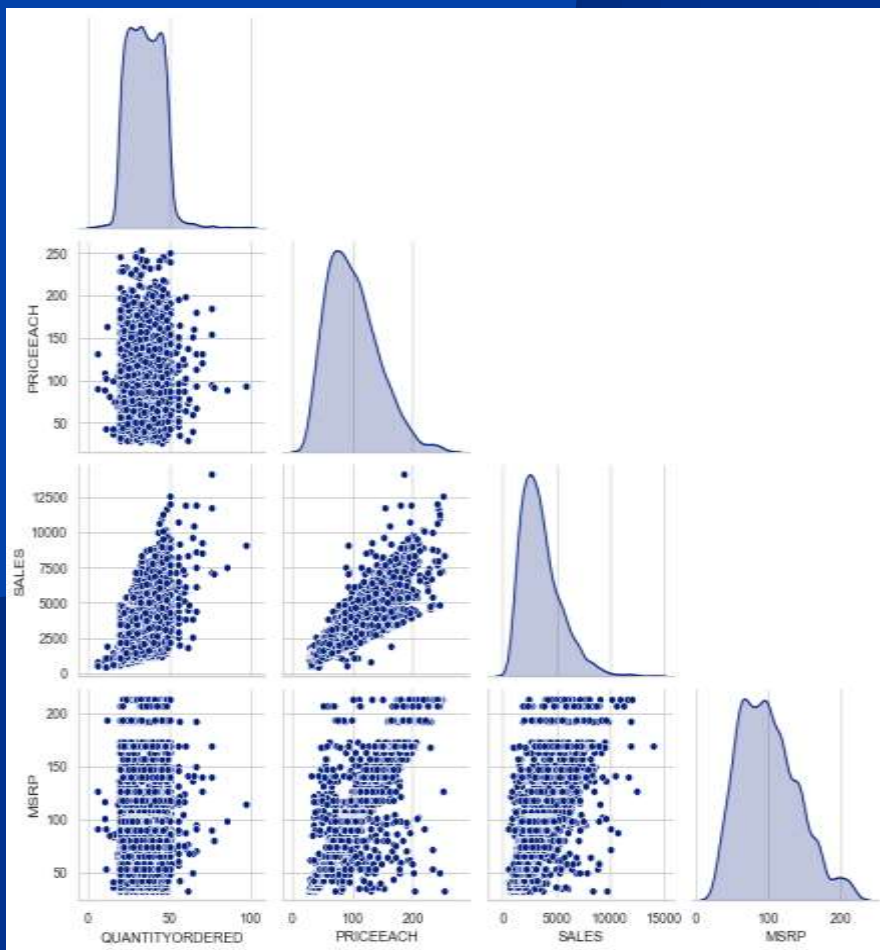


Country wise Sales Distribution

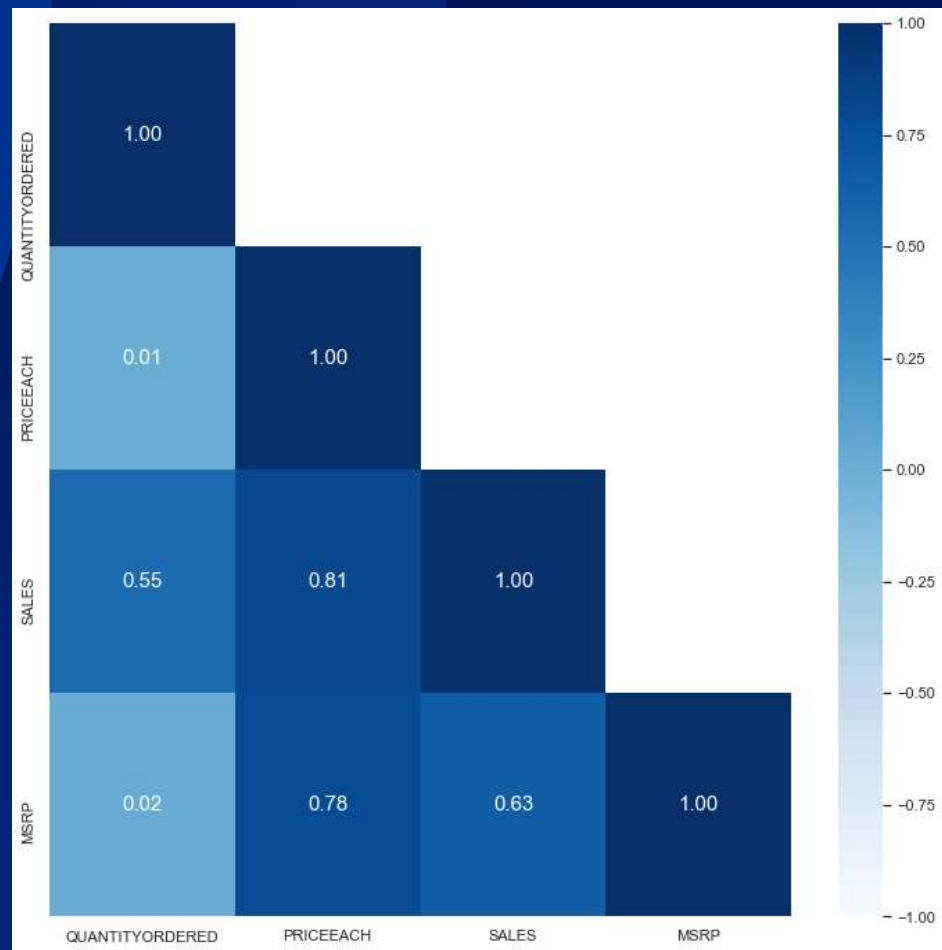


Customer wise Sales Distribution



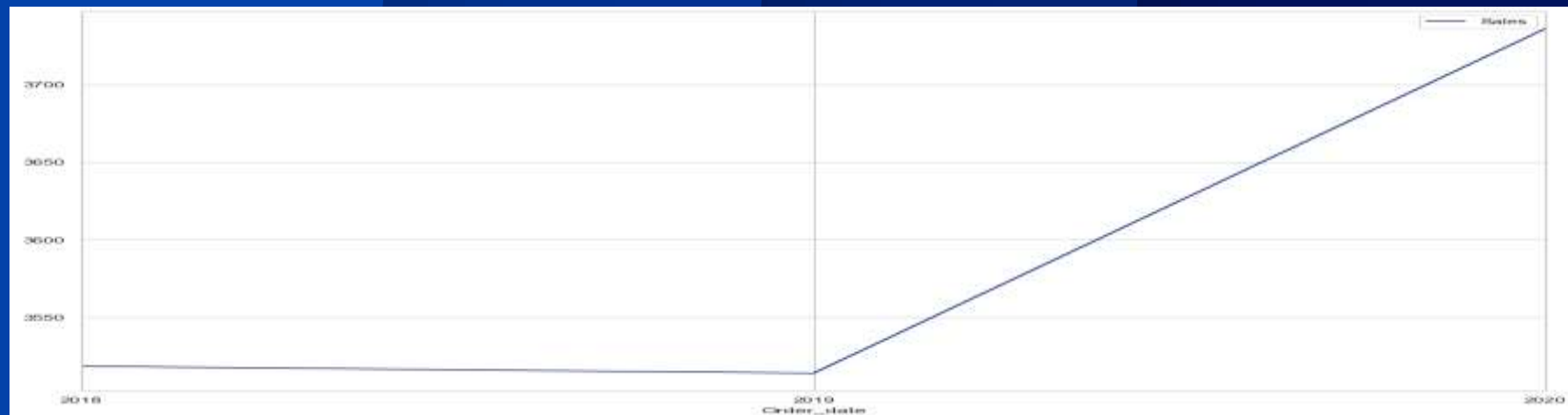


PAIRPLOT

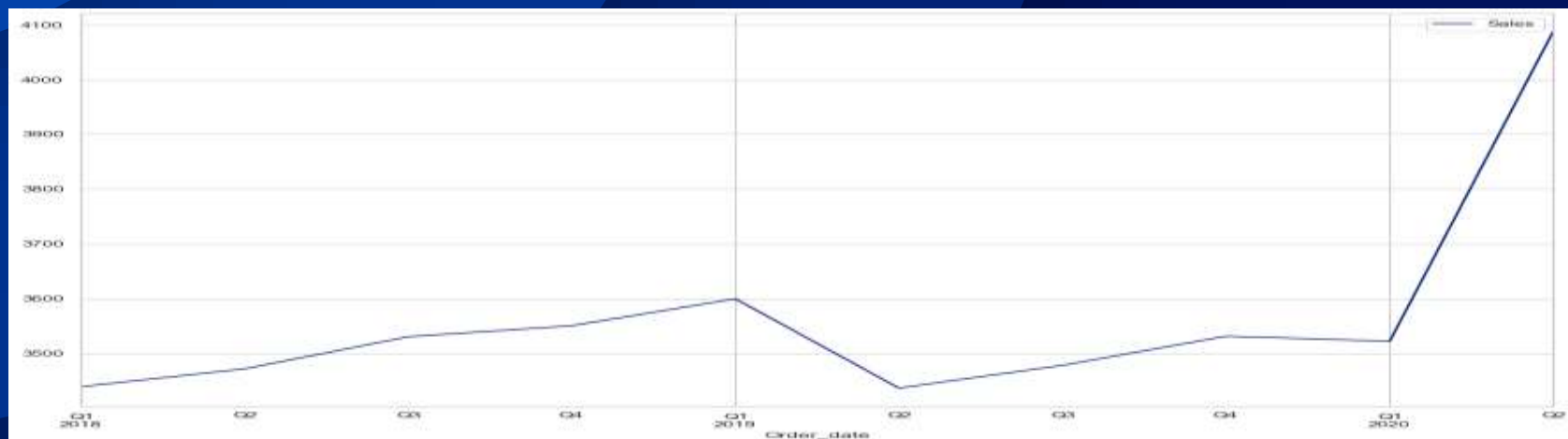


HEATMAP

Y
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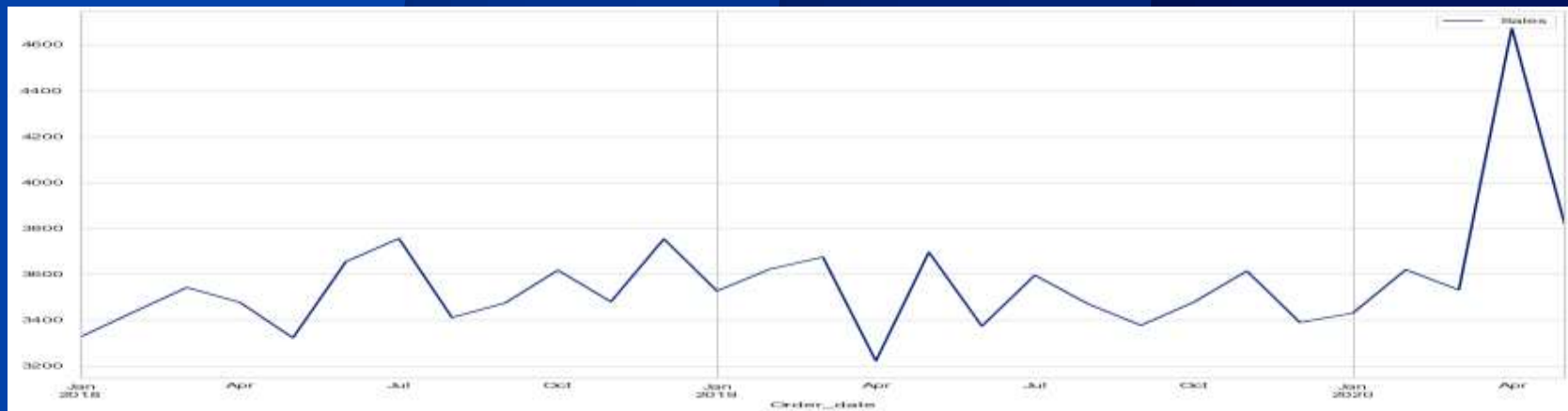


T
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N
D



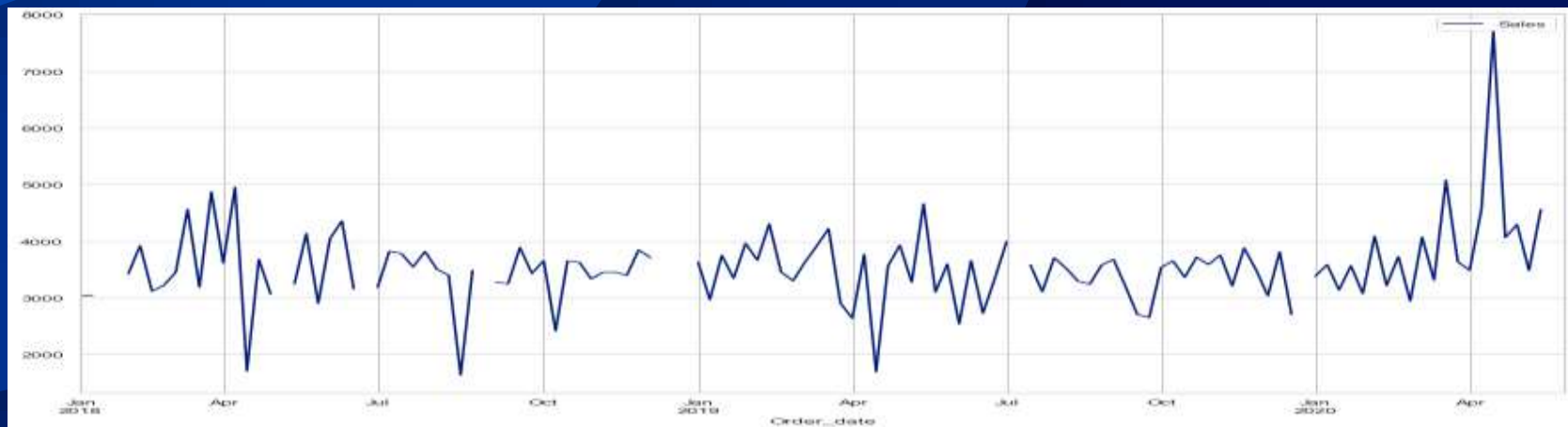
Q
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A
R
T
E
R
L
Y

MONTHLY



TREND

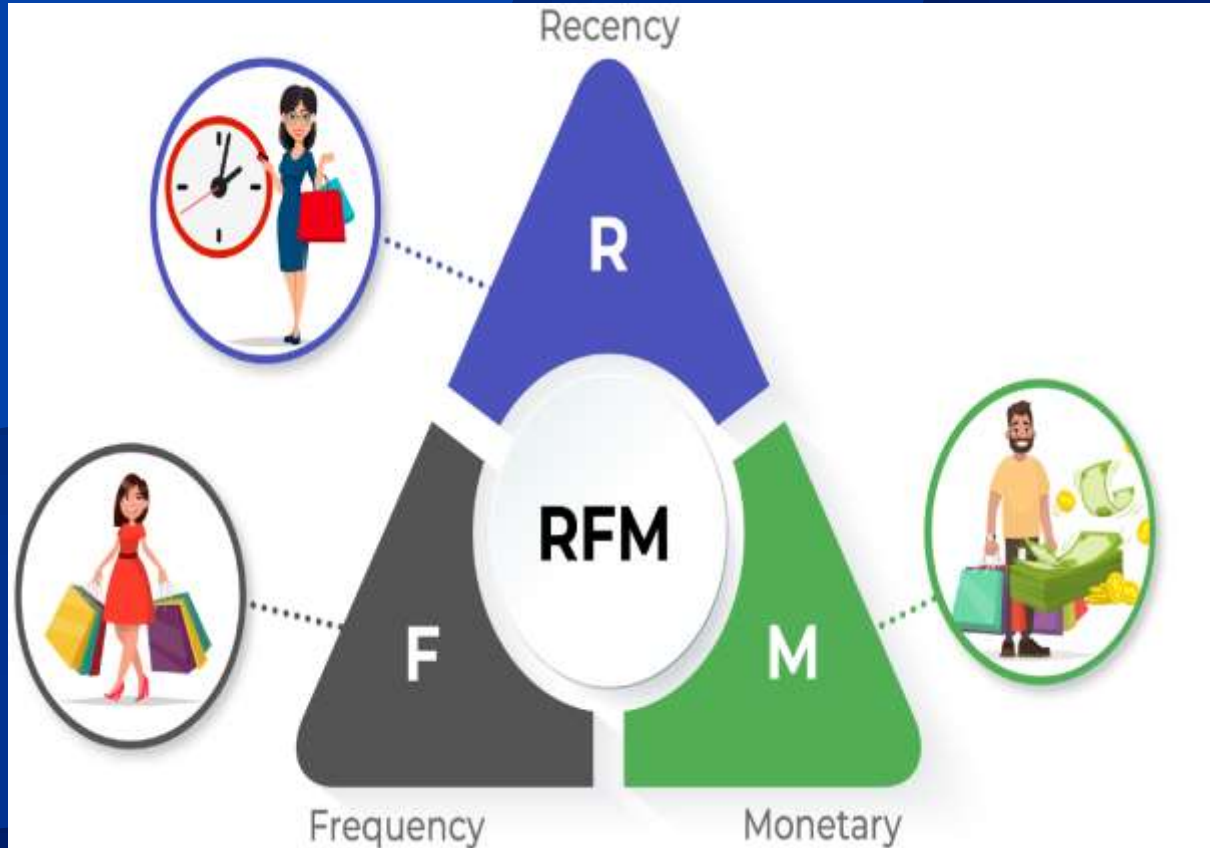
WEEKLY



INSIGHT FROM EDA

- **Total 2747 orders have been placed by 89 customers amounting USD 9760221.71** (currency assumed USD).
- From the histograms and boxplots, it is evident that the numerical variables are positively skewed and leptokurtic. The same fact of skewness is also evident from the Descriptive Statistics as mean > median in all cases.
- Some outliers are present in case of 'SALES' and 'PRICEEACH' while they are few in 'QUANTITYORDERED' and very few in the context of 'MSRP'. The few outliers in 'MSRP' may be due to high price suggested for expensive / premium products. **As the entries are seems to be genuine, we keep the entries as they are.**
- **Euro Shopping Channel** is the top customer in terms of both quantity & amount of orders.
- Parts for **Classic & Vintage cars** fetch the most sales revenue (around 58 %).
- More than 92 % orders have been **shipped**. Very less instances of cancellation and holding of orders.
- Most of the orders are of **medium deal size** (around 60%).
- **USA** alone accounts for more than 1/3rd of all sales, almost equal to the next 4 countries put together (Spain, France, Australia and UK). Top 6 countries (these 5 & Italy) fetches 75 % of the total sales.
- **50 % of the sales are ordered by Top 27 % customers** (24 out of 89).
- From pairplot, it is observed that **Sales revenue** shows a clear **and direct linear relationship** with actual Price of each item.
- The strong correlation between **Sales and Price of each item** can be validated with the Heatmap (corr = 0.81), much stronger than its correlation with MSRP or Quantity Ordered.
- Actual selling Price is also observed to be enjoying a **good correlation** with the Manufacturer Suggested Retail Price.
- Overall the data shows an **increasing trend**. The only exceptions are the sharp decline in Q2 over Q1 in 2019 Sales varies significantly a monthly basis. From monthly trend, it is clear that the decrease in Q2 2019 happened in the month of April (the lowest point in the entire dataset).
- The meteoric rise in 2020 happened during the month of April'20 may be happened due to **special event like expo / exhibition / carnival of vintage / classic cars.**

RFM ANALYSIS



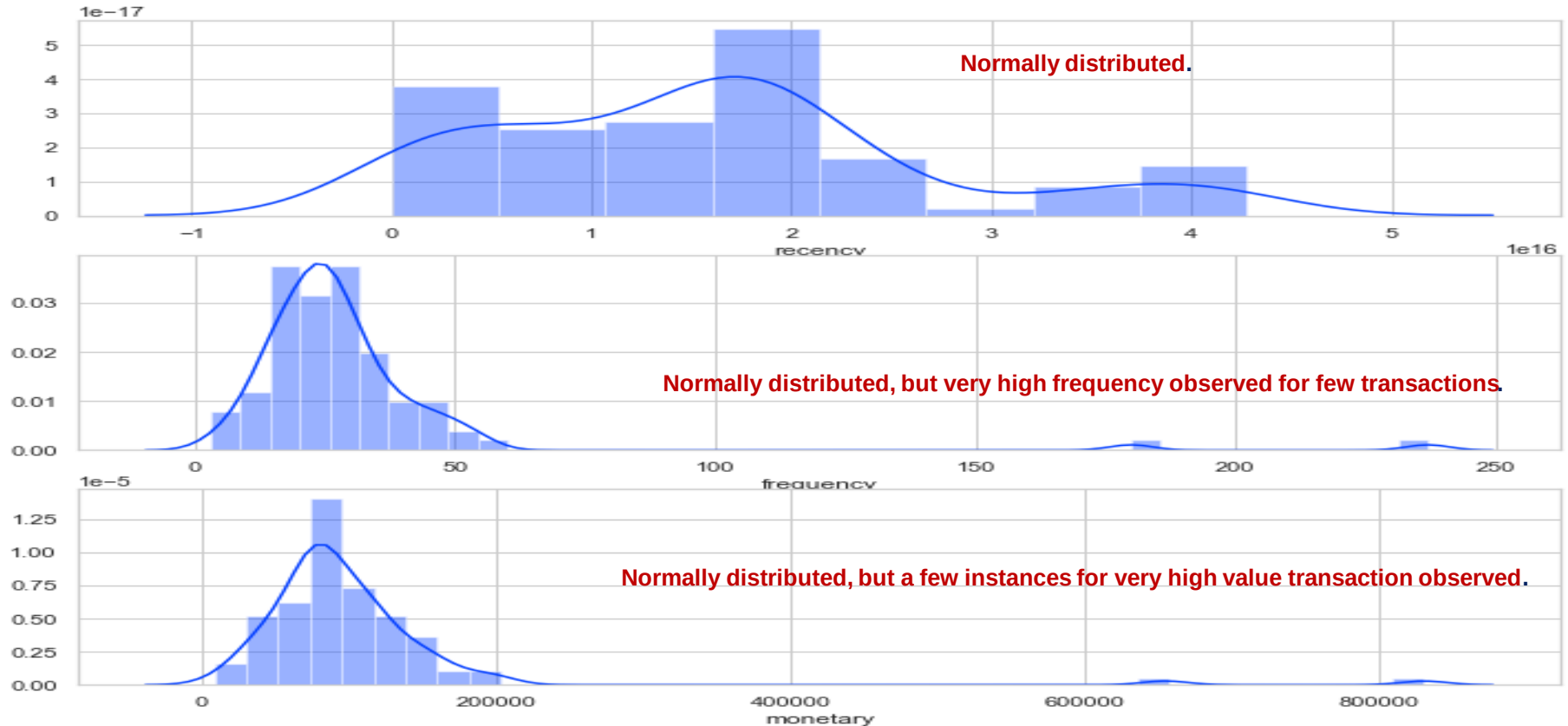
RFM factors illustrate these facts:

The more recent the purchase, the more responsive the customer is to promotions.

The more frequently the customer buys, the more engaged and satisfied they are.

Monetary value differentiates heavy spenders from low-value purchasers

RFM DISTRIBUTIONS



RFM ANALYSIS

TOOL USED : PYTHON (pandas, numpy, matplotlib, seaborn, rfm library called for)
EXCEL (pivot table)

PARAMETERS USED :

RECENCY : How recently a customer has made a purchase.

FREQUENCY : How often a customer makes a purchase.

MONETARY: How much money a customer spends on purchases

ASSUMPTIONS MADE :

1. The RFM analysis ranks each customer for each factor on a 1 to 4 scale (4 is highest & 1 is worst).
2. Basically, motor parts are mix of durable and nondurable items. The monetary value per transaction is normally high but frequency and recency is low in case of durable items whereas the order is opposite in case of nondurable So, we have decided to allot same weight to **monetary, recency and frequency**.
3. To arrive monetary value, we have not considered value of cancelled & disputed sales. The rows with those STATUS have been removed before RFM analysis.

Customer Segmentation using RFM analysis

The RFM analysis is based on the customers. So the final dataset is grouped by customer name to arrive RFM matrix.

Then we have applied RFM model with the appropriate weight as mentioned earlier.

Scoring norms used from 1 to 5 (5 as best & 1 as worst).

We have segmented the customers into 4 segments as 'Best', 'Loyal', 'Verge of Churning' & 'Lost'.

Best : With RFM Score of exactly 15 (Customers with all 5s)

Loyal : With RFM Score more than equals to 12 but less than 15 (Customers with at least 3 in each aspect / any equivalent combination)

Verge of Churning : With RFM Score more than 3 but less than 9 (Customers with at least more than 1 in any aspect)

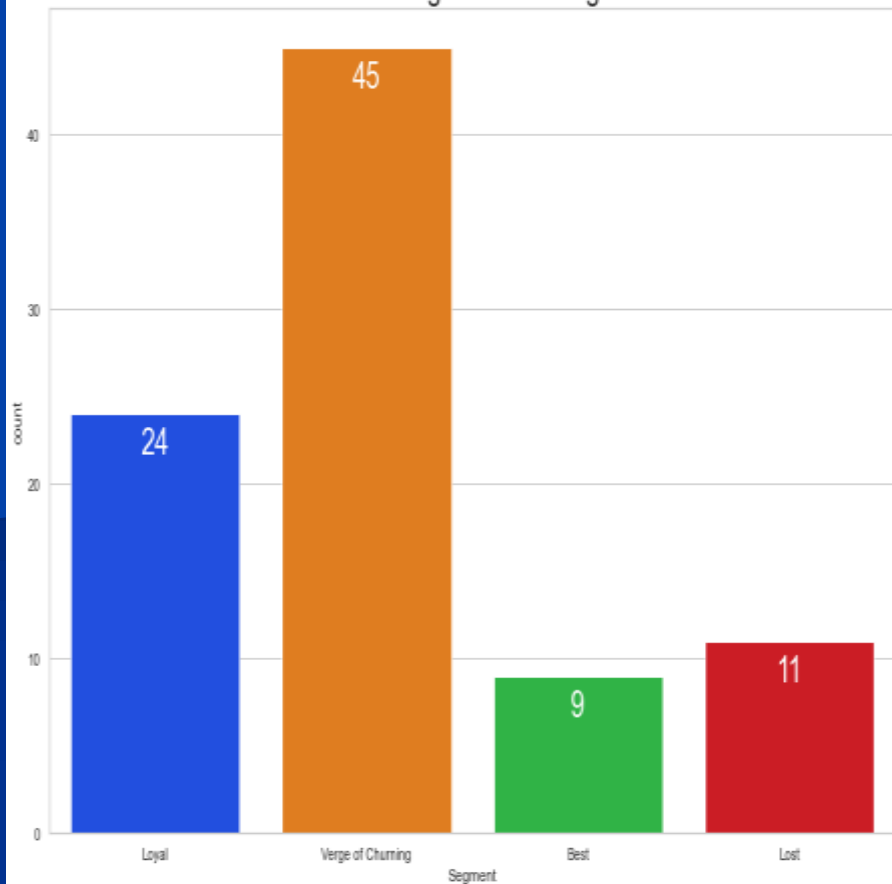
Lost : with RFM Score exactly 3 (Customers with all 1s)



RFM TABLE HEAD

Customer Name	recency	frequency	monetary	recency_scores	frequency_scores	monetary_scores	recency_weighted	frequency_weighted	monetary_weighted	rfm_score	Segment
AV Stores, Co.	196 days	51	157807.81	3	5	5	1.05	1.25	2.0	4.30	Loyal
Alpha Cognac	64 days	20	70488.44	4	2	2	1.40	0.50	0.8	2.70	Verge of Churning
Amica Models & Co.	265 days	26	94117.26	1	3	3	0.35	0.75	1.2	2.30	Verge of Churning
Anna's Decorations, Ltd	83 days	46	153996.13	4	5	5	1.40	1.25	2.0	4.65	Loyal
Atelier graphique	188 days	7	24179.96	3	1	1	1.05	0.25	0.4	1.70	Verge of Churning

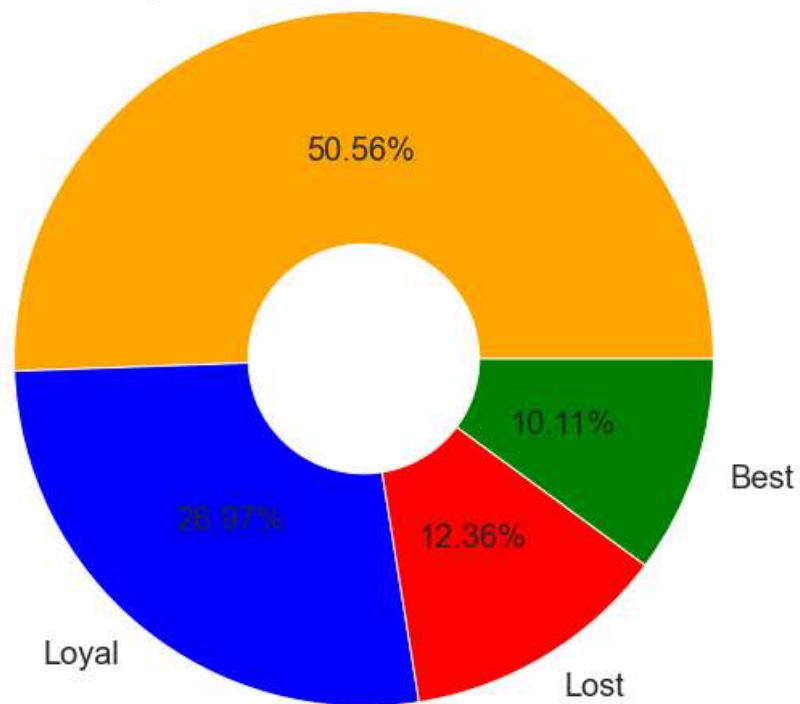
Bar Chart showing customer segmentation



Count wise Distribution

Donut Chart showing customer segmentation

Verge of Churning



Percentage wise Distribution

Number of Customers					
Recency Levels	Frequency Levels	Monetary Levels			
		4	3	2	1
4	4	9	1		
	3	2	1	1	
	2		1	5	
	1			1	2
3	4	4	1		
	3		6	1	
	2			4	
	1				6
2	4	5			
	3	1	6	1	
	2		2	4	1
	1			1	1
1	4	1			
	3		3	1	
	2		1	1	2
	1			2	11



Best Customers



Loyal Customers



Verge of Churning Customers



Lost Customers

Analyzing RFM Segmentation

- 'Best' Customers :** They bought most recently, most often, and are heavy spenders. Reward these customers. They can become early adopters for new products and will help any promotional campaign. They are the ambassadors.
- 'Loyal' Customers :** Potential Loyalists are recent customers with average frequency and who spent a good amount. Offer membership or loyalty programs or recommend related products to them and help them become 'Best' customers .
- 'Verge of Churning' Customers :** Customers who purchased less frequently and spent average amounts, but haven't purchased recently. Send them personalized reactivation campaigns to reconnect, and offer renewals and helpful products to encourage another purchase.
- 'Lost' Customers :** Customers who has visited long back and purchase for negligible amount. Should not invest on them and let them go. However, an internal deep analysis is required to find out what went wrong and avoid losing them to a competitor.

BEST CUSTOMERS

Euro Shopping Channel

L'ordine Souveniers

La Rochelle Gifts

Mini Gifts Distributors Ltd.

Salzburg Collectables

Souveniers And Things Co.

The Sharp Gifts Warehouse

LOYAL CUSTOMERS

Danish Wholesale Imports

Dragon Souveniers, Ltd.

Scandinavian Gift Ideas

Reims Collectables

Anna's Decorations, Ltd

UK Collectables, Ltd.

Tokyo Collectables, Ltd

VERGE OF CHURNING

Microscale Inc.

Iberia Gift Imports, Corp

Clover Collections, Co.

Diecast Collectables

Super Scale Inc.

Osaka Souveniers Co.

Daedalus Designs Imports

LOST CUSTOMERS

Bavarian Collectables Imports,
Co.

CAF Imports

Cambridge Collectables Co.

Double Decker Gift Stores, Ltd

Online Mini Collectables

Signal Collectibles Ltd.

West Coast Collectables Co.



Thank you

A presentation by
MAINAK
GHOSH